

Week 5- End to end ethical pipeline

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End-to-end ethical pipeline (extraction → cleaning → transformation → modeling)

Example : Student performance risk model (Education analytics study)

Goal: Predict which students may need support next week (early warning), using LMS activity + attendance + assessment history.

Step 1 — Extraction (ethical risks start here)

Sources: LMS clickstream (logins, resources viewed, time spent), Attendance system and Gradebook

Ethical pitfalls:

- Consent & purpose creep: Data collected for learning may be repurposed for discipline decisions.
- Sensitive proxies: “Time online” can be a proxy for socioeconomic status, job commitments, caregiving.
- Access control: Raw logs can expose highly personal behaviour.

Good practice

- Define purpose limitation: “support interventions only” (explicitly exclude wrong use).
- Use data minimization: extract aggregates (weekly activity counts) instead of raw URLs or page-level logs.
- Implement tiered access/role based access: raw logs restricted; feature tables accessible.

Is this a data sheet?

No: This is a data schema
section of data sheet

Data sheet represents what the
data *means*, how it was created,
and how it should *not* be used.

Datasheet is a structured
document that accompanies a
dataset, similar to:
a nutrition label for food,
a spec sheet for hardware.
It answers governance-critical
questions that raw data never
reveals.

Column name	Type	Description
student_id	String	Pseudonymized unique identifier
course_id	String	Course code
week_number	Integer	Academic week
attendance_sessions	Integer	Number of sessions attended
total_sessions	Integer	Number of scheduled sessions
lms_login_days	Integer	Days logged into LMS
lms_time_minutes	Integer	Total LMS time (minutes)
assignments_due	Integer	Assignments due this week
assignments_submitted	Integer	Assignments submitted
quiz_score_avg	Float	Average quiz score (0–100)
submission_late_count	Integer	Number of late submissions
device_type	String	Desktop / Mobile / Tablet
internet_access_flag	Boolean	Self-declared reliable access
week_start_date	Date	Start of week

Analytical ethics review of columns

Column	Ethical status
student_id	Sensitive → must be pseudonymized
course_id	Allowed

← Direct indicator

Use hashed student_id and keep re-identification keys in a separate secure system

Column	Risk	Notes
attendance_sessions	Low	Core academic signal
total_sessions	Low	Needed for normalization
assignments_submitted	Low	Outcome-related
quiz_score_avg	Medium	Can create pressure if misused

← Attendance and academic behaviour

These must not be used for punishment and use as support-only signals

Column	Risk	Why
lms_login_days	Medium	Proxy for time availability
lms_time_minutes	High	Proxy for SES, caregiving, work
submission_late_count	Medium	Can reflect constraints, not motivation

← Engagement data

- Aggregate weekly (not per-hour granularity)
- Never interpret in isolation
- Pair with attendance & submission completion

Engagement ≠ effort ≠ ability

Column	Risk	Handling
device_type	Medium	Proxy for income
internet_access_flag	High	Proxy for disadvantage



Sensitive proxies

- Do NOT use directly in modelling
- Use only for: Bias analysis, Missingness explanation, Fairness diagnostics

Ethically Transformed Dataset

Column name	Derived from	Ethical reasoning
student_id_hash	student_id	Privacy-preserving
course_id	course_id	Contextual
week_number	week_number	Temporal
attendance_rate	attendance_sessions / total_sessions	Normalized
assignment_completion_rate	submitted / due	Opportunity-aware
quiz_trend_3w	rolling avg	Avoids single-event bias
late_submission_flag	submission_late_count > 0	Binary → less punitive
engagement_bucket	lms_time_minutes	Low / Medium / High
missing_data_flag	any NA	Structural signal

Original for comparison

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Key take away-till now

- Need to remove Raw LMS minutes, Device type, Internet access flag as these becomes proxies for various other factors other than academics
- Ethical design often means removing information, not adding more.
- We need to create a Target Variable (label): Student support label

Target Variable (Label): Ethical Construction

Column	Description
needs_support_next_week	1 if ≥ 1 missed session AND ≥ 1 missed assignment next week

- Label reflects support need, not “student quality”
- Avoids moral judgment
- Ethical as it is not a ground truth of ability or effort

Ethical pipeline walkthrough

Revisiting STEP 1 — Extraction

- Sources: LMS (aggregated logs only), Attendance system, Gradebook
- Ethical risks: Consent ambiguity, Purpose creep, Exposure of sensitive behaviour
- Controls: Data minimization (weekly aggregates only), Purpose limitation (“support interventions only”) and Tiered/ role based access (raw logs restricted)

STEP 2 — Cleaning (accuracy + fairness)

- Common tasks: Remove duplicates, standardize IDs, time alignment , missing value handling
- Ethical angle
 - Missingness is often structural, not noise. Students with poor connectivity generate sparse logs.
 - If you “drop missing rows,” you might systematically remove disadvantaged students.
- Better approach
 - Create missingness indicators: missing_attendance_flag, missing_lms_flag
 - Use robust imputation strategies and track them in metadata.(ex: track missingness by course/week)

Ethical pipeline walkthrough...Contd

STEP 3 — Transformation (where bias often gets encoded)

- Feature engineering: Normalization, binning, rolling trends, removal of proxy heavy variables
 - Examples:
 - weekly_login_count
 - late_submission_rate
 - assessment_trend_last_3 (slope)
 - attendance_rate
- Bias risk addressed: Proxy discrimination, temporal leakage and overinterpretation of engagement
 - Examples:
 - “Engagement features” can punish neurodivergent students or those with external constraints.
 - Fairness review per feature: “Could this feature be a proxy for protected attributes?”, “Does it measure opportunity or behaviour shaped by constraints?”
 - Leakage risk: using future data (e.g., grades after the prediction week)
Better approach : Time-aware feature windows (only past data).

Ethical pipeline walkthrough...Contd

STEP 4 — Modeling (performance + interpretability + constraints)

- Model choices: Interpretable baselines first: logistic regression, monotonic Gradient Boost Methods.
- Keep a complex model only if it improves outcomes without harming subgroup error rates.
- Deployment guardrails always needed
- Output risk bands (Low/Med/High) with explanation rather than a single score used for punishment.
- Human-in-the-loop: advisors see explanations + recommended support actions.

STEP 5 — Access Control

(Role-Based + Purpose-Based Access Control)

Role	Access
Advisor	Risk band + explanations
Instructor	Aggregate trends only
Admin	Audit logs only
Student	Own data only

Final STEP: ACTION

Support flag triggered because:

- Attendance rate decreased this week
- Assignment completion dropped compared to the last 3 weeks
- This does not reflect ability or effort.
- Note that there is no mention of: LMS time, Device used or Internet access (which can create purpose creep)

Completing the Ethical Pipeline: Explainability through Metadata & Documentation

In an ethical data pipeline, explainability is not an afterthought and not limited to model interpretability alone. It is embedded throughout the pipeline via metadata, documentation, and governance artifacts.

Think of explainability as **two parallel deliverables**:

1. Modal level explainability: Why did the model make this prediction?
 - Local explanations (per student)
 - Global explanations (overall behavior)
 - Constrained to non-sensitive, non-proxy features
2. System-level explainability: How was this data produced, transformed, governed, and used?
 - Dataset documentation
 - Feature documentation
 - Dataset contract
 - Lineage and provenance
 - Access control and auditability

System level explainability: Governance Artifacts Embedded Across the Pipeline

1. **Datasheet for the Dataset:** *(Explains the data itself)*

The datasheet accompanies the dataset and answers governance-critical questions before any modeling occurs.

- Motivation: early academic support only
- Composition: student-week records, aggregated signals
- Collection process: LMS, attendance, gradebook
- Proxy risks: engagement variables may reflect SES, caregiving, work
- Allowed uses: support
- Prohibited uses: grading, ranking, discipline
- Retention & access rules: role-based, time-limited

Without a datasheet, downstream researchers or practitioners may:

- Misinterpret engagement as effort
- Repurpose data for punitive decisions
- Ignore structural bias encoded in proxies

System level explainability

2. **Feature Documentation:** *(Explains how each variable should be interpreted)*

Each engineered feature in the pipeline is documented before it is allowed into modeling.

For every feature, document:

- Definition (exact formula)
- Temporal window (e.g., last 7 days, 30 days)
- Refresh rate
- Known proxy risks/ biases
- Allowed and disallowed uses(Policy)

Example feature doc snippet:

- weekly_login_count
 - Definition: Number of distinct LMS login days in the past 7 days
 - Bias /proxy risk: May reflect connectivity, work schedule or caregiving (SES)
 - Usage constraint: Must not be used alone for intervention decisions
 - Monitoring: Track subgroup mean shifts; missingness correlation

System level explainability

3. **Data Contracts** (*Explains what the pipeline expects from source systems*)

- Example: attendance_table contract
 - Primary key: student_id, date
 - Allowed values: attendance_status $\in$ {present, absent, excused}
 - Freshness: updated daily by 2am
 - Null rules: student_id must never be null
 - Drift monitors:
 - weekly % absent
 - missing rate by course
- Without data contracts:
 - Schema changes can silently bias models
 - Missing data patterns may shift unfairly
 - Fairness and performance degrade unnoticed

System level explainability

4. **Dataset Lineage & Provenance:** *(Explains how the dataset came to exist)*

The pipeline records:

- Source systems used (LMS, attendance, gradebook)
- Queries and filters applied
- Joins performed
- Aggregation logic
- Dataset versions and timestamps

This enables answers to questions such as:

- “Which data influenced this output?”
- “Was the data stale or missing?”
- “What transformations were applied?”

Lineage is essential for:

- Accountability
- Appeals and dispute resolution
- Regulatory and institutional audits

System level explainability

5. Access control and Auditability: *(Explains how the system was actually used)*

Role	Permitted Access	Purpose Constraint
Advisor	Student-level risk band + explanations	Academic support only
Instructor	Aggregated class-level trends	Course improvement
Administrator	Metadata, logs, audit summaries	Governance & compliance
Student	Own records and explanations	Transparency & trust

Access control prevents: Purpose creep, Unauthorized surveillance, Misuse of predictive outputs.

Audit logs:

- Model version used
- Feature set version
- Threshold / policy version
- Identity and role of the accessor and timestamp of access
- Whether an intervention was triggered (privacy-aware)

Audit logs enable: Accountability for decisions, Detection of misuse or policy violations, Response to student appeals or disputes and Institutional and regulatory review

Model card for data sheet

Model Card: *(Explains the trained model and its behaviour)*

The model card documents the model as a governed artifact.

- Intended use: support
- Explicit exclusions: no disciplinary decisions, admission decision
- Training data reference (linked datasheet)
- Metrics: precision, recall, ROC AUC
- Subgroup checks (where legally/ethically allowed)
- Explainability method (e.g., SHAP with constraints)
- Known limitations and bias observations
- Monitoring and retraining plan

A model without a model card:

- Cannot be responsibly deployed
- Cannot be audited
- Cannot be ethically defended

The model card explains *what the model does and must not do*.

End-to-End Explainability: From Data to Decisions

Pipeline stage	Explainability artifact
Extraction	Datasheet + data contracts
Cleaning	Data quality reports + missingness metadata
Transformation	Feature documentation + lineage
Modeling	Model card + interpretability methods
Deployment	Audit logs + access control
Monitoring	Drift reports + retraining records

Ethical pipelines do not eliminate risk—they make risk explicit, bounded, and accountable.

Adult Census Income from UCI Machine Learning Repository (example dataset with good governance)

Adult - UCI Machine Learning Repository

1. Clear provenance – Accountability and traceability
 - who collected the data
 - for what original purpose
 - how it was curated and redistributed
2. Explicit purpose (income prediction research)- purpose limitation
3. Well-documented features (meaning, type, known limitations), Sensitive attributes are explicitly labeled, not hidden.
This enables
 - informed feature selection
 - bias analysis
 - explainable modeling
4. No direct identifiers (Names, addresses, SSNs are removed, Individuals cannot be re-identified directly)
5. Stable versioning
 - The dataset does not silently change
 - Research results are reproducible
 - Models trained today can be audited later
6. Governance principles demonstrated
 - Data minimization
 - Purpose specification
 - Documentation & lineage
 - Controlled reuse

MIMIC-III / MIMIC-IV-(example dataset with good governance)

Medical Information Mart for Intensive Care MIMIC-III / MIMIC-IV- <https://physionet.org/content/mimiciv/3.1/>

It contains data for over 65,000 patients admitted to an ICU in Boston and over 200,000 patients admitted to the emergency department

1. Governance designed *before* data release (not retrofitted)
 - Data sharing was planned as part of the clinical data lifecycle, not as an afterthought.
 - Governance requirements shaped how data were collected, stored, de-identified, and released.
2. Institutional ethical oversight (IRB-approved release)- Legitimacy & ethical accountability (Data release is approved by IRB)
3. Mandatory training before access
 - Users must complete: human-subjects research training (e.g., CITI) data ethics and privacy modules
 - Access is logged and tied to a real identity.
 - This prevents naive misuse, accidental re-identification and unethical experimentation
4. Strong de-identification (Privacy by design)
 - Direct identifiers removed
 - Dates shifted consistently per patient
 - Rare events generalized or suppressed
 - De-identification aligned with HIPAA Safe Harbor
5. Clear data use agreement which explicitly states the permitted uses (research, education), prohibited uses (re-identification, commercial misuse) and consequences of violation for auditability
6. Consent and ethical oversight
7. Clear documentation & data lineage